

HOMework 1

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 09/08/2023

Problem 1. Consider the problem of evaluating a polynomial $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ of degree n at some specified value of x .

- As a function of n , how many operations (i.e. additions and multiplications) are performed in the following algorithm to evaluate p at the value x ?

Algorithm 1: The obvious algorithm to evaluate a polynomial.

Input: $n \geq 0, x \in \mathbb{R}, a = [a_0, \dots, a_n]$

Output: $v = a_0 + a_1 x + \dots + a_n x^n$

```
v ← a[0]
for i = 1, ..., n do
    y ← 1
    for j = 1, ..., i do
        y ← yx
    end
    v ← v + a[i]y
end
return v
```

- Another option for polynomial evaluation is Horner's method, which is based on the observation that

$$\begin{aligned} p(x) &= a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \\ &= (a_n x + a_{n-1}) x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 \\ &= ((a_n x + a_{n-1}) x + a_{n-2}) x^{n-2} + a_{n-3} x^{n-3} + \dots + a_1 x + a_0 \\ &\vdots \\ &= (((\dots ((a_n x + a_{n-1}) x + a_{n-2}) x + \dots + x) a_2 + a_1) x + a_0. \end{aligned}$$

As a function of n , how many operations (i.e. additions and multiplications) are performed in the following algorithm implementing Horner's method to evaluate p at the value x ?

Algorithm 2: An algorithm to evaluate a polynomial using Horner's method.

Input: $n \geq 0, x \in \mathbb{R}, a = [a_0, \dots, a_n]$

Output: $v = a_0 + a_1 x + \dots + a_n x^n$

```
v ← a[n]
for i = 1, ..., n do
    v ← v * x + a[n - i]
end
return v
```

Problem 2. Give a simple example where relative error is a more suitable measure than absolute error, and another example where the absolute error measure is more suitable.

Problem 3. Explain briefly why accumulation of roundoff errors is inevitable when arithmetic operations are performed in a floating point system.

Problem 4. Consider a floating point system with a given base b and precision t and let $\eta = \frac{1}{2}b^{1-t}$ be the rounding unit. By definition, $f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$. Accordingly, one might attempt to approximate $f'(1)$ via $f'(1) \approx \frac{f(1+h) - f(1)}{h}$ using some fixed small value of h . What happens when $0 < h < \eta$?

Problem 5. Consider a floating point system with a given base b and precision t and let $\eta = \frac{1}{2}b^{1-t}$ be the rounding unit. Explain why $\text{fl}(1+x) = 1$ if $0 < x < \eta$, assuming that the fl operation rounds to the nearest floating point number and that $1+x$ is evaluated exactly.